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STUDENT NAME: _____

TEACHER: _____

THE HILLS GRAMMAR SCHOOL

TASK 4 Trial Examination 2015

YEAR 12

MATHEMATICS EXTENSION 2

Time Allowed: Three hours (plus five minutes reading time)

Weighting: 40%

Outcomes: E1, E2, E3, E4, E5, E6, E7, E8, E9

Instructions:

- Approved calculators may be used
- Attempt all questions
- Start all questions on a new sheet of paper
- The marks for each question are indicated on the examination
- Show all necessary working

MCQ	Question 11	Question 12	Question 13	Question 14	Question 15	Question 16	TOTAL
10	15	15	15	15	15	15	100

Section 1 Multiple Choice (10 Marks)

1 The $\int \frac{x}{\sqrt{9-4x^2}} dx$ is:

(A) $-\frac{\sqrt{9-4x^2}}{4} + c$

(B) $\frac{\sqrt{9-4x^2}}{4} + c$

(C) $-\frac{3\sqrt{9-4x^2}}{2} + c$

(D) $\frac{3\sqrt{9-4x^2}}{2} + c$

2 The $\int \frac{1}{x^2-6x+13} dx$ is:

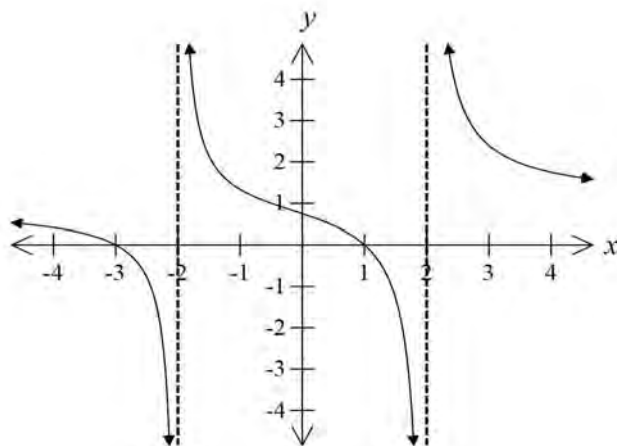
(A) $\tan^{-1} \frac{x-3}{2} + c$

(B) $\frac{1}{2} \tan^{-1}(x-3) + c$

(C) $\frac{1}{2} \tan^{-1} \frac{x-3}{2} + c$

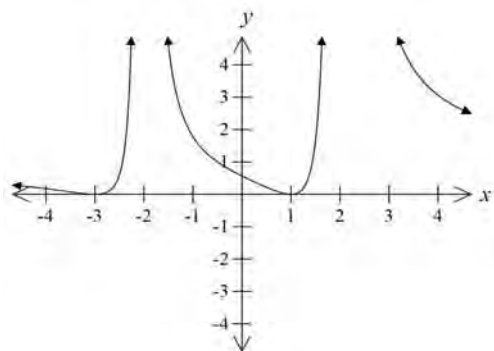
(D) $\frac{1}{4} \tan^{-1} \frac{x-3}{4} + c$

- 3 The diagram shows the graph of the function $y = f(x)$.

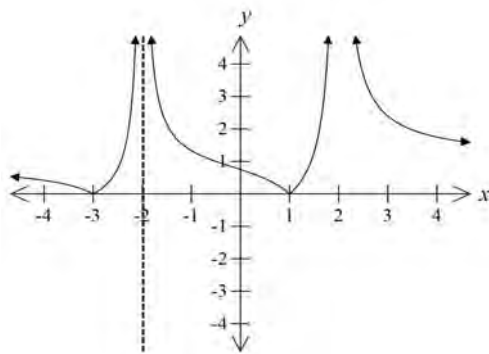


Which of the following is the graph of $y = |f(x)|$?

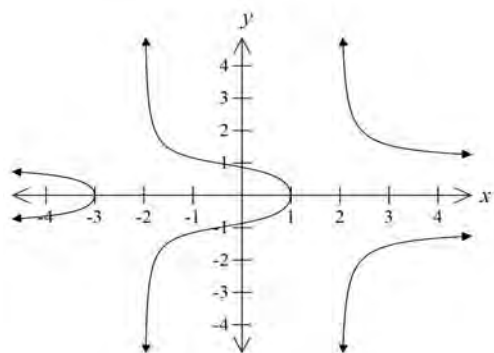
(A)



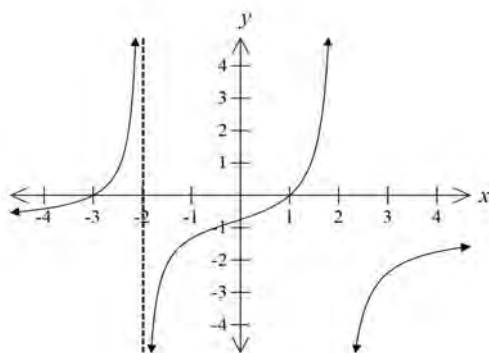
(B)



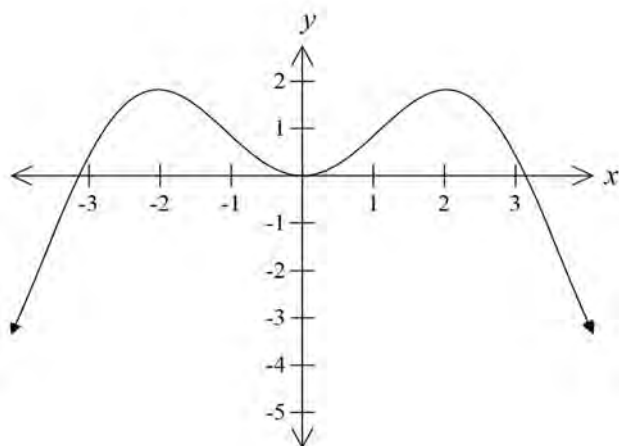
(C)



(D)

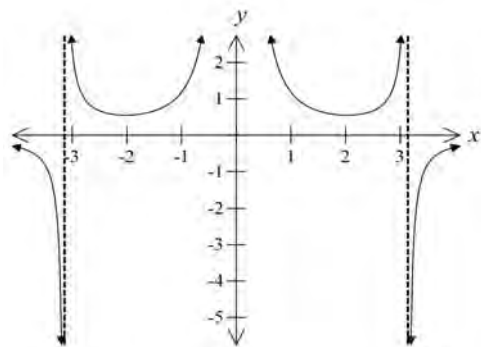


- 4 The diagram shows the graph of the function $y = f(x)$.

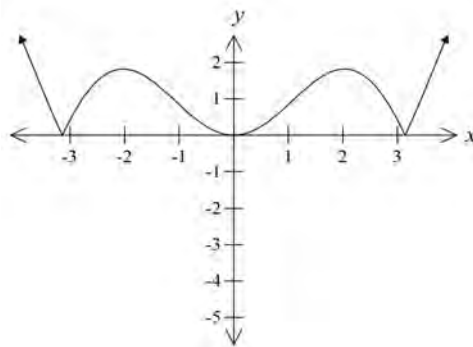


Which of the following is the graph of $y = \frac{1}{f(x)}$?

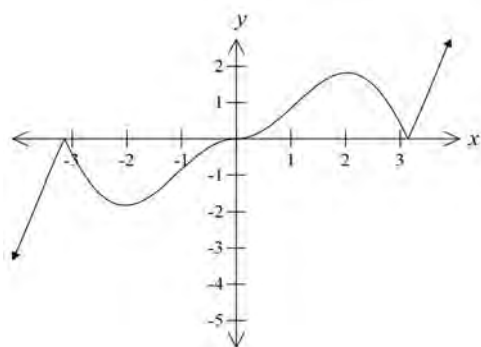
(A)



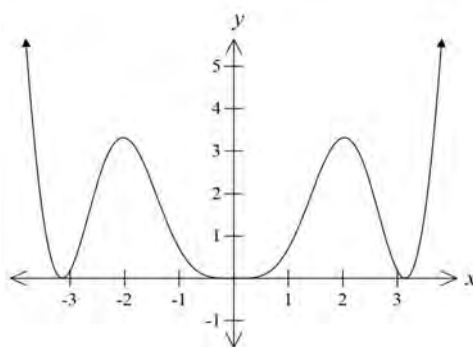
(B)



(C)



(D)



5 Let $z = 2 + i$ and $w = 1 - i$. What is the value of $3z + iw$?

(A) $5 - 4i$

(B) $5 + 4i$

(C) $7 + 4i$

(D) $7 - 4i$

6 It is given that $3 + i$ is a root of $P(z) = z^3 + az^2 + bz + 10$ where a and b are real numbers. Which expression factorises $P(z)$ over the real numbers?

(A) $(z - 1)(z^2 + 6z - 10)$

(B) $(z - 1)(z^2 - 6z - 10)$

(C) $(z + 1)(z^2 + 6z + 10)$

(D) $(z + 1)(z^2 - 6z + 10)$

7 For the ellipse with the equation $\frac{x^2}{4} + \frac{y^2}{3} = 1$. What is the eccentricity?

(A) $\frac{1}{4}$

(B) $\frac{1}{2}$

(C) $\frac{3}{4}$

(D) $\frac{9}{16}$

8 Consider the hyperbola with the equation $\frac{x^2}{4} - \frac{y^2}{3} = 1$.

What are the coordinates of the vertices of the hyperbola?

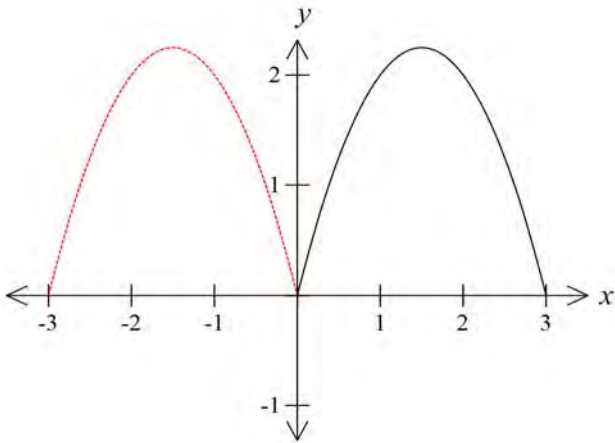
(A) $(0, \pm 2)$

(B) $(\pm 2, 0)$

(C) $(0, \pm 4)$

(D) $(\pm 4, 0)$

- 9 The area between the curve $y = 3x - x^2$, the x -axis, $x = 0$ and $x = 3$, is rotated about the y -axis to form a solid.



What is the volume of this solid?

- (A) $\frac{9\pi}{4}$ cubic units
- (B) $\frac{9\pi}{2}$ cubic units
- (C) $\frac{27\pi}{4}$ cubic units
- (D) $\frac{27\pi}{2}$ cubic units
- 10 A particle of mass m is moving in a straight line under the action of a force, $F = \frac{m}{x^3}(6 - 10x)$. Which of the following is an expression for its velocity in any position, if the particle starts from rest at $x = 1$?

- (A) $v = \pm \frac{1}{x} \sqrt{(-3 + 10x - 7x^2)}$
- (B) $v = \pm x \sqrt{(-3 + 10x - 7x^2)}$
- (C) $v = \pm \frac{1}{x} \sqrt{2(-3 + 10x - 7x^2)}$
- (D) $v = \pm x \sqrt{2(-3 + 10x - 7x^2)}$

Section 2

Marks

Question 11 (15 marks) Use a SEPARATE writing booklet.

(a) Let $z = 2\left(\cos\frac{\pi}{3} + i\sin\frac{\pi}{3}\right)$ and $\omega = \sqrt{3} + i$.

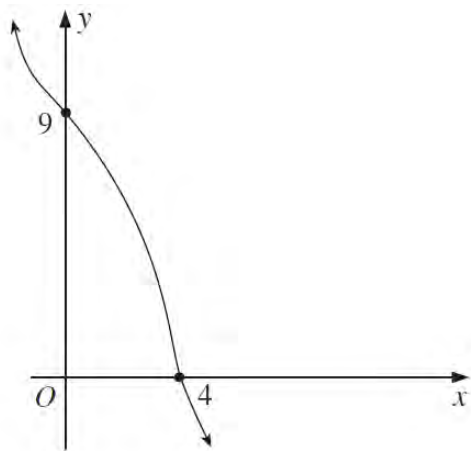
(i) Express ω in modulus-argument form. 1

(ii) Hence, or otherwise, express $z^3\omega$ in modulus-argument form. 2

(b) Sketch the region in the complex plane where the inequalities $|z + \bar{z}| \leq 1$ and $|z - i| \leq 1$ hold simultaneously. 2

(c) Evaluate $\int_0^2 te^{-t} dt$. 3

(d) The diagram shows the graph of the (decreasing) function $y = f(x)$.



Draw separate one-third page sketches of the graphs of the following:

(i) $y = |f(x)|$. 1

(ii) $y = \frac{1}{f(x)}$. 2

(iii) $y^2 = f(x)$. 2

(iv) $y = f^{-1}(x)$. 2

Question 12 (15 marks) Use a SEPARATE writing booklet.

(a) (i) Find the square roots of $3 + 4i$. 3

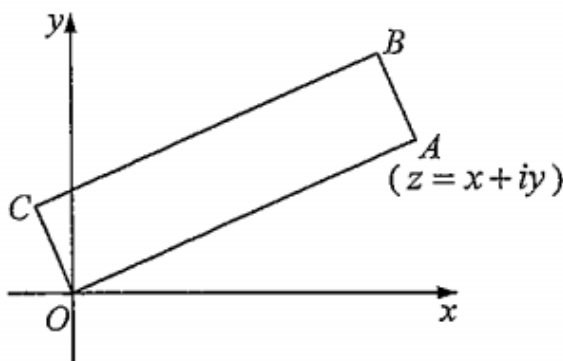
(ii) Hence, or otherwise, solve the equation $z^2 + iz - 1 - i = 0$. 2

(b) Use the substitution $t = \tan \frac{\theta}{2}$ to show that $\int_{\frac{\pi}{2}}^{\frac{2\pi}{3}} \frac{d\theta}{\sin \theta} = \frac{1}{2} \ln 3$. 3

(c) (i) Given that $\frac{16x-43}{(x-3)^2(x+2)}$ can be written as $\frac{a}{(x-3)^2} + \frac{b}{x-3} + \frac{c}{x+2}$ where a, b and c are real numbers, find a, b and c . 3

(ii) Hence find $\int \frac{16x-43}{(x-3)^2(x+2)} dx$. 2

(d) In the Argand diagram below, $OABC$ is a rectangle. O is the origin and the distance OA is four times the distance AB . The vertex A is represented by the complex number $z = x + iy$.



Find an expression for the complex number that represents the vertex B .
Leave your answer in the form $a + ib$. 2

Question 13 (15 marks) Use a SEPARATE writing booklet.

- (a) The equation $4x^3 - 27x + k = 0$ has a double root. Find the possible values of k . **2**
- (b) Let α, β and γ be the roots of the equation $x^3 - 5x^2 + 5 = 0$.
- (i) Find a polynomial equation with integer coefficients whose roots are $\alpha - 1, \beta - 1$ and $\gamma - 1$. **2**
- (ii) Find a polynomial equation with integer coefficients whose roots are α^2, β^2 and γ^2 . **2**
- (iii) Find the value of $\alpha^3 + \beta^3 + \gamma^3$. **2**
- (c) (i) Let $a > 0$. Find the points where the line $y = ax$ and the curve $y = x(x - a)$ intersect. **1**
- (ii) Let R be the region in the plane for which $x(x - a) \leq y \leq ax$. Sketch R . **1**
- (iii) A solid is formed by rotating the region R about the line $x = -2a$. Use the method of cylindrical shells to find the volume of the solid. **5**

Question 14 (15 marks) Use a SEPARATE writing booklet.

- (a) (i) Find all the 5th roots of -1 in modulus-argument form. 2
- (ii) Sketch the 5th roots of -1 on an Argand diagram. 1

(b) For each integer $n \geq 0$, let

$$I_n = \int_0^1 x^{2n+1} e^{x^2} dx.$$

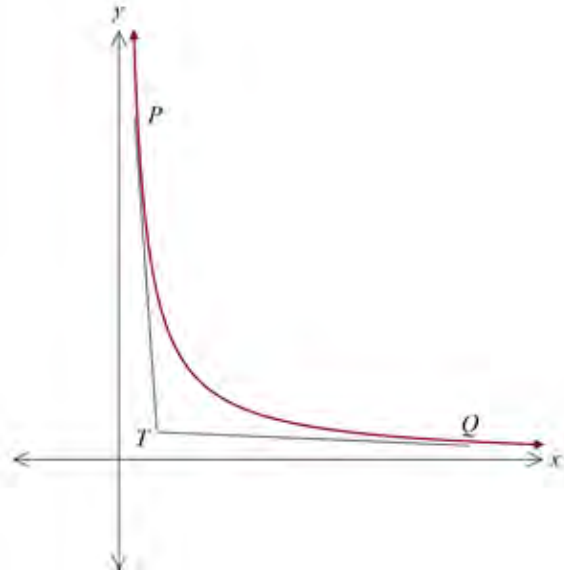
- (i) Show that for $n \geq 1$, $I_n = \frac{e}{2} - nI_{n-1}$ 2
- (ii) Hence, or otherwise, calculate I_2 . 2

(c) If $5x^2 - y^2 + 4xy = 18$ defines a set of points:

- (i) Using implicit differentiation show that it has no stationary points. 2
- (ii) Find the vertical tangents. 2
- (iii) Find any intercepts. 1
- (iv) Find the oblique asymptotes. 2
- (v) Sketch the curve. 1

Question 15 (15 marks) Use a SEPARATE writing booklet.

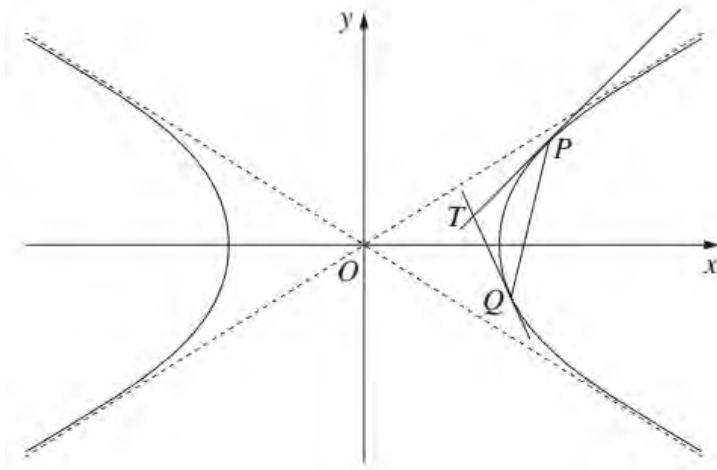
- (a) The points $P\left(cp, \frac{c}{p}\right)$ and $Q\left(cq, \frac{c}{q}\right)$, $p \neq q$, lie on the same branch of the hyperbola $xy = c^2$. The tangents at P and Q meet at the point T .



Find the equation of the tangent to the hyperbola at Q ?

2

- (b) The points at $P(x_1, y_1)$ and $Q(x_2, y_2)$ lie on the same branch of the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$.



The tangents at P and Q meet at $T(x_0, y_0)$.

- (i) Show that the equation of the tangent at P is $\frac{xx_1}{a^2} - \frac{yy_1}{b^2} = 1$. 2
- (ii) Hence show that the chord of contact, PQ , has equation $\frac{xx_0}{a^2} - \frac{yy_0}{b^2} = 1$. 2
- (iii) The chord PQ passes through the focus $S(ae, 0)$, where e is the eccentricity of the hyperbola. Prove that T lies on the directrix of the hyperbola. 1

- (c) In an alien universe, the gravitational attraction between two bodies is proportional to x^{-3} , where x is the distance between their centres.

A particle is projected upward from the surface of a planet with velocity u at time $t = 0$. Its distance x from the centre of the planet satisfies the equation $\ddot{x} = -\frac{k}{x^3}$.

- (i) Show that $k = gR^3$, where g is the magnitude of the acceleration due to gravity at the surface of the planet and R is the radius of the planet. 1

- (ii) Show that v , the velocity of the particle, is given by $v^2 = \frac{gR^3}{x^2} - (gR - u^2)$. 3

- (iii) It can be shown that $x = \sqrt{R^2 + 2uRt - (gR - u^2)t^2}$. (Do NOT prove this.)

Show that if $u \geq \sqrt{gR}$ the particle will not return to the planet. 2

- (iv) If $u < \sqrt{gR}$ the particle reaches a point whose distance from the centre of the planet is D , and then falls back.

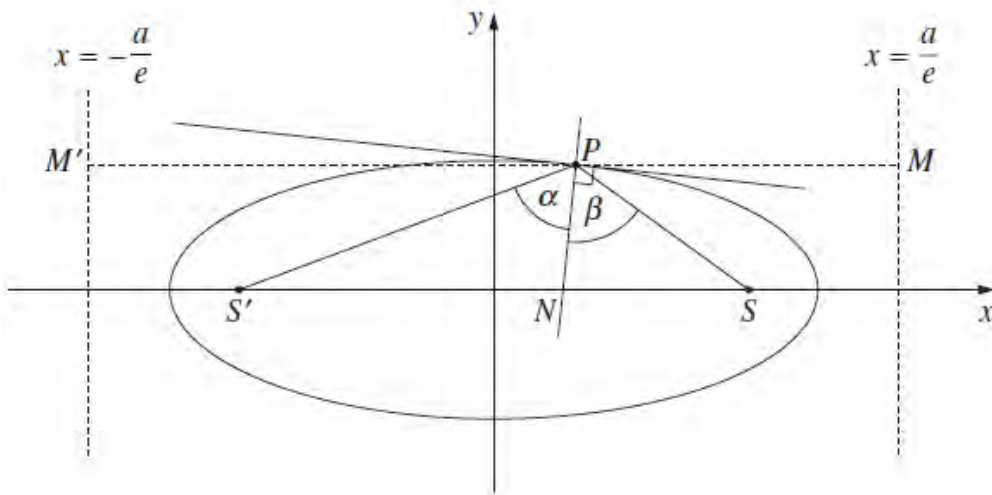
- (1) Use the formula in part (ii) to find D in terms of u , R and g . 1

- (2) Use the formula in part (iii) to find the time taken for the particle to return to the surface of the planet in terms of u , R and g . 1

Question 16 (15 marks) Use a SEPARATE writing booklet.

- (a) A flywheel of radius 30cm makes 30 revolutions per second. Find the velocity and acceleration of a point on the rim. 2

- (b) The ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ has foci $S(ae, 0)$ and $S'(-ae, 0)$ where e is the eccentricity, with corresponding directrices $x = \frac{a}{e}$ and $x = -\frac{a}{e}$. The point $P(x_0, y_0)$ is on the ellipse. The points where the horizontal line through P meets the directrices are M and M' , as shown in the diagram below.



- (i) Show that the equation of the normal to the ellipse at the point P is

$$y - y_0 = \frac{a^2 y_0}{b^2 x_0} (x - x_0). \quad 2$$

- (ii) The normal at P meets the x -axis at N . Show that N has coordinates $(e^2 x_0, 0)$. 2

- (iii) Using the focus-directrix definition of an ellipse, or otherwise,

show that $\frac{PS}{PS'} = \frac{NS}{NS'}$ 2

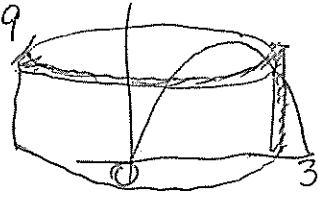
- (iv) Let $\alpha = \angle S'PN$ and $\beta = \angle NPS$. By applying the sine rule to $\angle S'PN$ and to $\angle NPS$, show that $\alpha = \beta$. 2

- (c) The gravitational force between two objects of masses m and M placed at a distance x apart is proportional to their masses and inversely proportional to the square of their distance, ie $F \propto \frac{Mm}{x^2}$. A satellite is launched so that it orbits the earth once a day. Take gravity at the earth's surface, $g = 9.8ms^{-2}$ and the radius of the earth, $R = 6400km$.

- (i) Find the angular velocity of the satellite. **1**
- (ii) Show that the centripetal force of the satellite $mr\omega^2$ is equal to $\frac{(6.4 \times 10^6)^2 \times 9.8m}{x^2}$. **2**
- (iii) Hence find the height of the satellite. **1**
- (iv) Find the linear velocity of the satellite. **1**

END OF ASSESSMENT

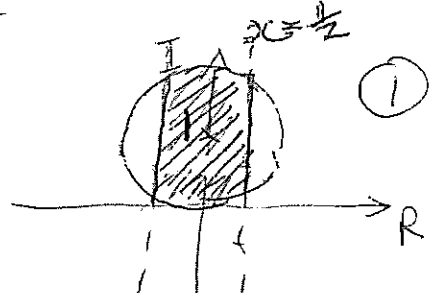
Suggested Solutions, Marking Scheme and Markers' comments

Suggested solution(s) SECTION 1 (MCQ)	comments
1/ $\int x(9-4x^2)^{\frac{5}{2}} dx = \frac{1}{-8 \times \frac{1}{2}} (9-4x^2)^{\frac{1}{2}}$ $= -\frac{1}{4} \sqrt{9-4x^2}$ part(A)	
2/ $\int \frac{1}{x^2-6x+13} dx = \int \frac{1}{x^2-6x+9+4} dx$ $= \int \frac{1}{(x-3)^2+4} dx$ part(C) $= \frac{1}{2} \tan^{-1}\left(\frac{x-3}{2}\right) + C.$	
3/ part(B)	
4/ part(A)	
5/ $z=2+i$ $w=1-i$ $3z+4w = 6+3i+i-i^2 = 7+4i$ part(C)	
6/ $P(z) = z^3 + az^2 + bz + 10$ $\text{prod} = -10$ $\alpha = 3+i$ $\beta = 3-i$ $\alpha\beta = 10$ $\alpha\beta\gamma = -10 \Rightarrow \gamma = -1$ $\therefore P(z) = (z+1)(z^2-6z+10)$ part(D)	
7/ $b^2 = a^2(1-e^2)$ $e < 1$ ellipse $e^2 = \frac{a^2-b^2}{a^2} = \frac{4-3}{4}$ part(B) $e = \frac{1}{2}$	
8/ $\frac{x^2}{4} - \frac{y^2}{3}$ when $y=0$ $x = \pm 2$ vertices $(\pm 2, 0)$ part(B)	
9/  Shells $dV = 2\pi x h dx$ $= 2\pi x(3x-x^2) dx$ $V = 2\pi \int_0^3 3x^2 - x^3 dx$ $= 2\pi \left[x^3 - \frac{x^4}{4} \right]_0^3$ $= 2\pi \left(27 - \frac{81}{4} \right) = \frac{27\pi}{2}$ part(D)	

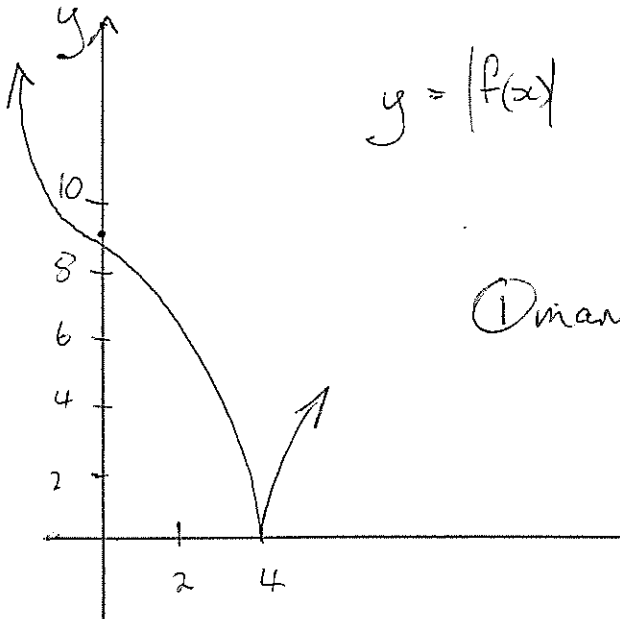
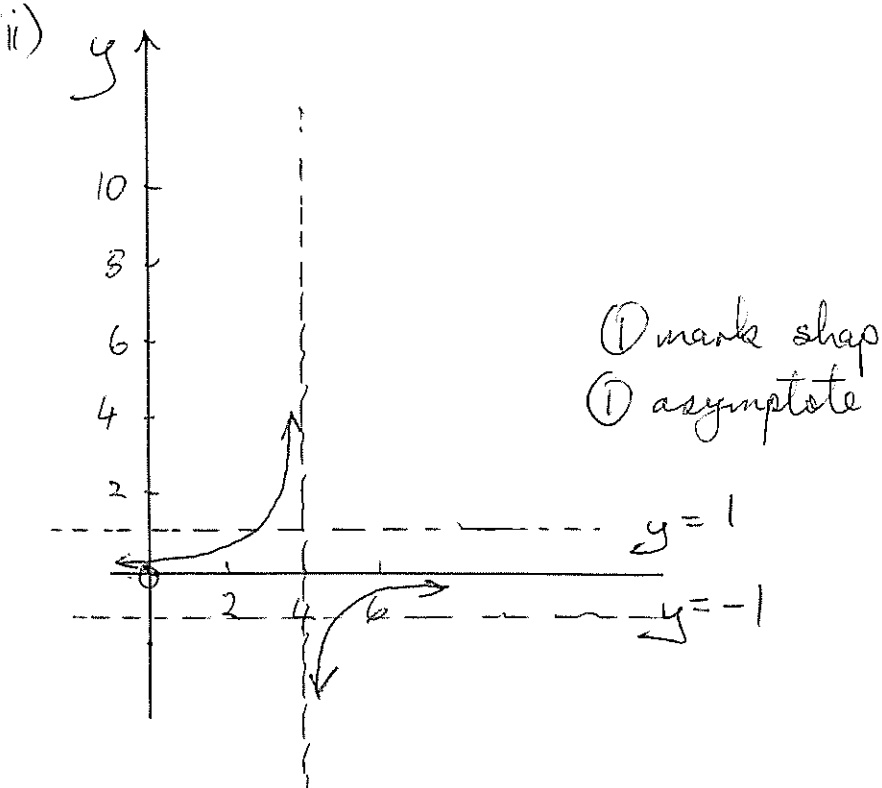
Suggested Solutions, Marking Scheme and Markers' comments

<u>Suggested solution(s)</u> SECT 1	<u>comments</u>
10/ $F = \frac{m}{x^3} (6 - 10x)$ $\ddot{x} = \frac{6 - 10x}{x^3} = 6x^{-3} - 10x^{-2}$ $\frac{1}{2}v^2 = \frac{6x^{-2}}{-2} + \frac{10x^{-1}}{1} + c$ $v^2 = -3x^{-2} + 20x^{-1} + 2c$ when $x = 1$, $v = 0$ $0 = -6 + 20 + 2c \Rightarrow 2c = -14$ $v^2 = -6x^{-2} + 20x^{-1} - 14$ $= \frac{1}{x^2} (-6 + 20x - 14x^2)$ $v = \pm \frac{1}{x} \sqrt{2(-3 + 10x - 7x^2)}$ part (c)	

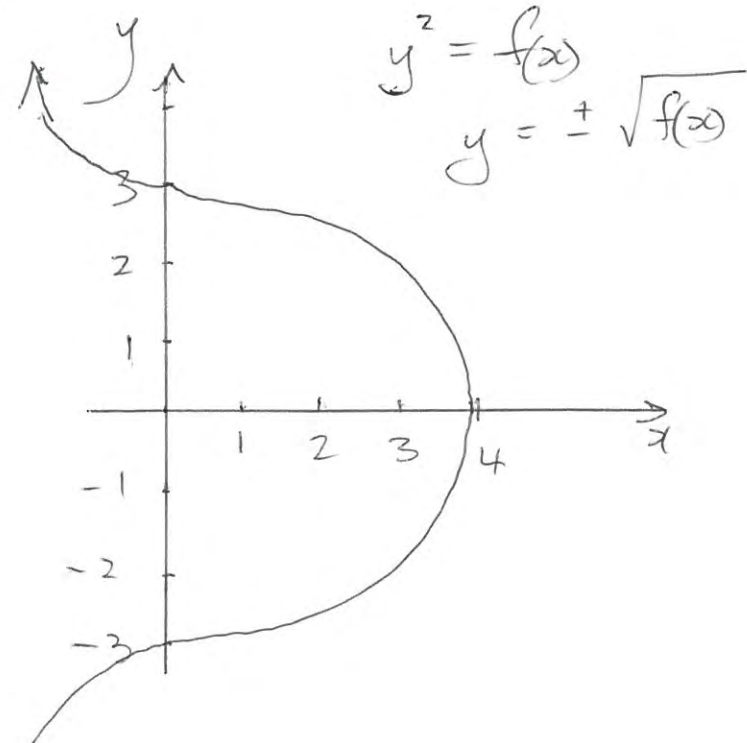
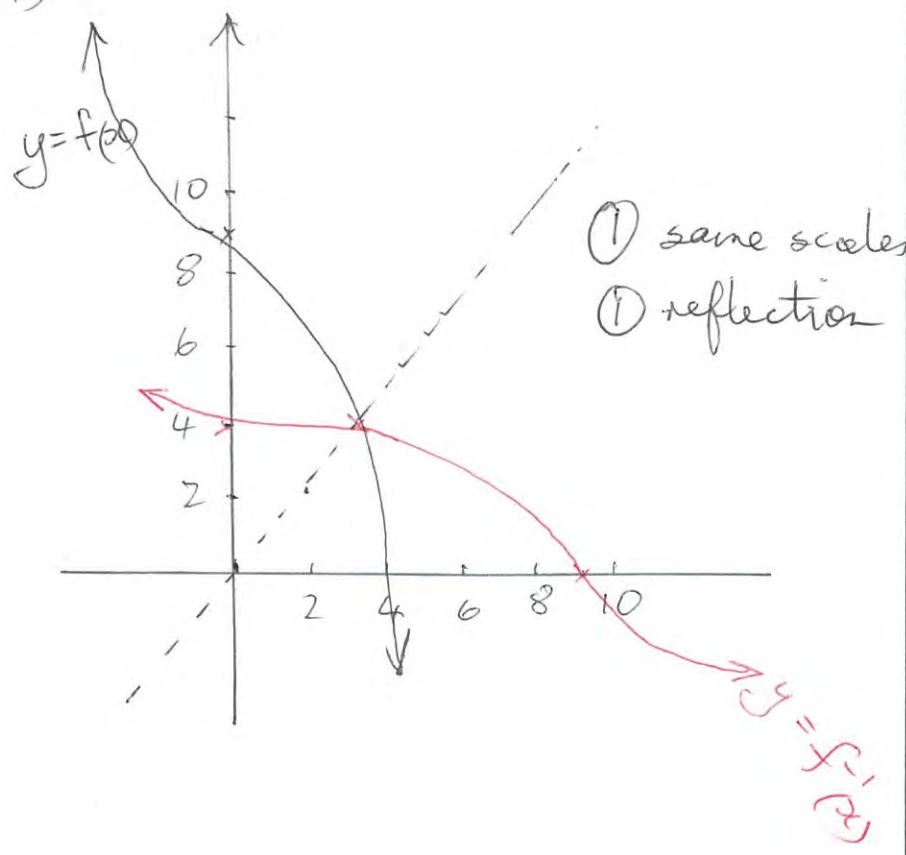
Suggested Solutions, Marking Scheme and Markers' comments

Suggested solution(s) Question 11	comments
(a) (i) $w = 2\left(\frac{\sqrt{3}}{2} + \frac{i}{2}\right) = 2\left(\cos\frac{\pi}{6} + i\sin\frac{\pi}{6}\right)$ $= 2\cos\frac{\pi}{6}$ ① mark. (ii) $z^3 = 2^3 \cos 3 \times \frac{\pi}{3} = 8\cos\pi$ ① Mark $z^3 w = 8\cos\pi \cdot 2\cos\frac{\pi}{6}$ $= 16\cos\frac{\pi}{6}$ ① Mark $= 16\cos\left(-\frac{5\pi}{6}\right) + i\sin\left(-\frac{5\pi}{6}\right)$	
(b) $z + \bar{z} \leq 1$ $2x \leq 1$ $-\frac{1}{2} \leq x \leq \frac{1}{2}$ ① Mark  ① Mark.	
(c) $\int_0^2 t e^{-t} dt$ ① mark. $\begin{cases} u=t & v=e^{-t} \\ u'=1 & v=-e^{-t} \end{cases}$ $= \left[-te^{-t}\right]_0^2 + \int_0^2 e^{-t} dt$ ① Mark $= -2e^{-2} + \left[-e^{-t}\right]_0^2$ $= -2e^{-2} - e^{-2} + 1$ $= 1 - 3e^{-2}$ ① Mark.	

Suggested Solutions, Marking Scheme and Markers' comments

Suggested solution(s)	Question 11	comments
(d)	 $y = f(x)$ ① mark	
(i)	 ① mark shape ① asymptote	

Suggested Solutions, Marking Scheme and Markers' comments

<u>Suggested solution(s)</u>	<u>comments</u>
iii)  $y^2 = f(x)$ $y = \pm \sqrt{f(x)}$ iv)  $y = f(x)$ $y = f^{-1}(x)$ ① same scales ① reflection	

Suggested Solutions, Marking Scheme and Markers' comments

Suggested solution(s) Question 12	comments
(a)(i) $a + ib = \sqrt{3+4i}$ $a^2 - b^2 + 2abi = 3 + 4i$ (1 mark) $a^2 - b^2 = 3 \quad ab = 2$ $a = \pm 2 \quad b = \pm 1$ (1 mark) $\therefore$ square roots are $\pm(2+i)$ (ii) $z^2 + iz - 1 - i = 0$ $z = \frac{-i \pm \sqrt{i^2 + 4(1+i)}}{2}$ (1 mark) $= \frac{-i \pm \sqrt{-1+4+4i}}{2}$ $= \frac{-i \pm \sqrt{3+4i}}{2}$ $= \frac{-i \pm (2+i)}{2}$ $= \frac{2}{2} \text{ or } \frac{-2-2i}{2}$ $= 1, -1-i$ (1 mark)	
(b) $t = \tan \frac{\theta}{2}$ $dt = \frac{1}{2} \sec^2 \frac{\theta}{2} d\theta$ $dt = \frac{1}{2} (1 + \tan^2 \frac{\theta}{2}) d\theta$ $d\theta = \frac{2}{1+t^2} dt$ deriv 2 when $\theta = \frac{2\pi}{3} \quad t = \sqrt{3}$ limits $\quad \quad \quad = \frac{\pi}{2} \quad t = 1$ (1 mark) $\int_{\frac{\pi}{2}}^{\frac{2\pi}{3}} \frac{2}{1+t^2} d\theta = \int_1^{\sqrt{3}} \frac{1}{t} dt = [\ln t]_1^{\sqrt{3}}$ $\quad \quad \quad = \ln \sqrt{3} = \frac{1}{2} \ln 3$ (1 mark) (1 mark)	

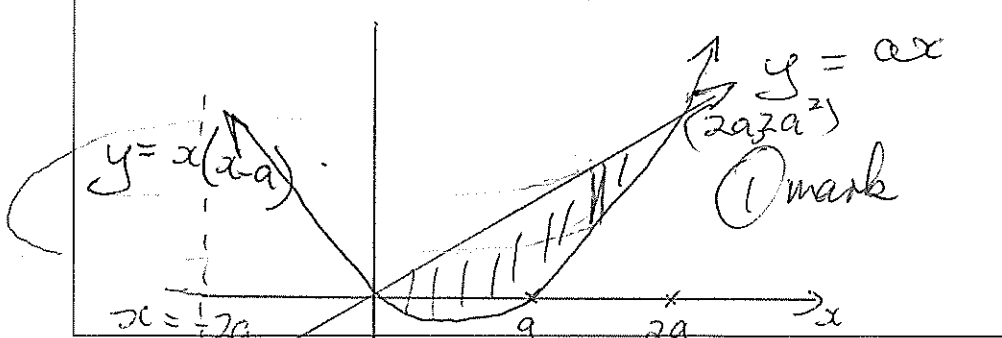
Suggested Solutions, Marking Scheme and Markers' comments

Suggested solution(s) Question 12	comments
(c)(i) $\frac{16x-43}{(x-3)^2(x+2)} = \frac{a}{(x-3)^2} + \frac{b}{(x-3)} + \frac{c}{x+2}$ $16x-43 \equiv a(x+2) + b(x-3)(x+2) + c(x-3)^2$ let $x=3$ $48-43 = 5a \Rightarrow a=1$ ① mark let $x=-2$ $-32-43 = 25c \Rightarrow c=-3$ ① mark let $x=0$ $-43 = 2a - 6b + 9c$ $-43 = 2 - 6b - 27$ $6b = 18 \Rightarrow b=3$ ① mark (ii) $\therefore \int \frac{16x-43}{(x-3)^2(x+2)} dx = \int \frac{1}{(x-3)^2} + \frac{3}{x-3} - \frac{3}{x+2} dx$ $= -\frac{1}{(x-3)} + 3\ln(x-3) - 3\ln(x+2)$ $= \frac{-1}{x-3} + 3\ln\left(\frac{x-3}{x+2}\right) + C$ ① mark ① mark.	
(d) $\vec{AB} = \frac{1}{4} i \vec{AO}$ $\vec{OB} = \vec{OA} + \vec{AB} \quad \text{① mark}$ $= x+iy + \frac{i}{4}(x+iy)$ $= \left(x - \frac{1}{4}y\right) + i\left(y + \frac{1}{4}x\right)$ ① mark	

Suggested Solutions, Marking Scheme and Markers' comments

Suggested solution(s) Question 13	comments
(a) $P(x) = 4x^3 - 27x + k$ $P'(x) = 12x^2 - 27$ for double root $12x^2 - 27 = 0$ $x^2 = \frac{27}{12} = \frac{9}{4}$ $x = \pm \frac{3}{2}$ (1 mark) when $x = \frac{3}{2}$ $P(x) = 4 \times \frac{27}{8} - 27 \times \frac{3}{2} + k$ $= -27 \times \frac{3}{2} + k$ $k = 27$ $x = -\frac{3}{2}$ $P(x) = -4 \times \frac{27}{8} + \frac{27 \times 3}{2} + k$ $= \frac{3}{2} \times 27 + k$ $k = -27$ (1 mark) (b) $x^3 - 5x^2 + 5 = 0$ (i) roots α, β, γ Equate with roots $\alpha-1, \beta-1, \gamma-1$ $y = x-1 \Rightarrow x = 1+y$ (1 mark) $(1+y)^3 - 5(1+y)^2 + 5 = 0$ $y^3 + 3y^2 + 3y + 1 - 5 - 10y - 5y^2 + 5 = 0$ $y^3 - 2y^2 - 7y + 1 = 0$ (1 mark) (ii) for roots $\alpha^2, \beta^2, \gamma^2$ $y = x^2 \Rightarrow x = \sqrt{y}$ (1 mark) $(y^{\frac{1}{2}})^3 - 5(y^{\frac{1}{2}})^2 + 5 = 0$ $y^{\frac{3}{2}} - 5y + 5 = 0$ $y^{\frac{3}{2}} = 5y - 5$ $y^3 = 25y^2 - 50y + 25$ $y^3 - 25y^2 + 50y - 25 = 0$ (1 mark)	

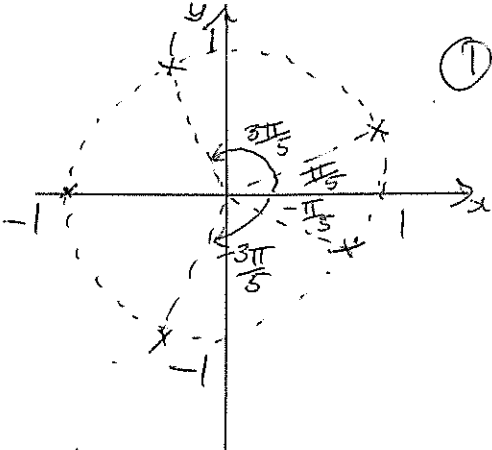
Suggested Solutions, Marking Scheme and Markers' comments

Suggested solution(s) <i>Quest 13</i>	comments
(b)(iii) for roots $\alpha^3 + \beta^3 + \gamma^3$ $\alpha^3 = 5\alpha^2 + 5 = 0$ $\beta^3 - 5\beta^2 + 5 = 0$ $\gamma^3 - 5\gamma^2 + 5 = 0$ $\alpha^3 + \beta^3 + \gamma^3 - 5(\alpha^2 + \beta^2 + \gamma^2) + 15 = 0$ $\alpha^3 + \beta^3 + \gamma^3 = 5(\alpha^2 + \beta^2 + \gamma^2) - 15$ ① mark $\alpha^2 + \beta^2 + \gamma^2 = (\alpha + \beta + \gamma)^2 - 2(\alpha\beta + \alpha\gamma + \beta\gamma)$ $= 5^2 - 2 \times 0$ $= 25$ $\therefore \alpha^3 + \beta^3 + \gamma^3 = 5 \times 25 - 15$ $= 110 \quad \text{① mark}$	
(c) (i) $y = ax$ ① $y = x(x-a) = x^2 - ax$ ② equating ① & ② $x^2 = 2ax = 0$ $x(x-2a) = 0$ $x = 0, 2a$ when $x = 0, y = 0$ $x = 2a, y = 2a^2$ pts $(0, 0)$ $(2a, 2a^2)$ ① mark  ① mark	

Suggested Solutions, Marking Scheme and Markers' comments

<u>Suggested solution(s)</u> Question 13	<u>comments</u>
(c) $\delta V = \underbrace{2\pi r h}_{\text{curved surface area of cylinder}} \cdot \underbrace{\delta x}_{\text{length}} \quad \text{1 mark for 1}$ $\delta V = 2\pi(2a+x)(y_1-y_2)\delta x \quad \text{1 mark}$ $= 2\pi(2a+x)(ax-x^2+ax)\delta x$ $V = 2\pi \int_0^{2a} (2a+x)(2ax-x^2) dx \quad \text{1 mark}$ $= 2\pi \int_0^{2a} 4a^2x + 2ax^2 - 2ax^2 - x^3 dx$ $= 2\pi \left[24a^2 \frac{x^2}{2} - \frac{x^4}{4} \right]_0^{2a} \quad \text{1 mark}$ $= 2\pi \left[2a^2 \times 4a^2 - \frac{16a^4}{4} \right]$ $= 2\pi (8a^4 - 4a^4)$ $= 8\pi a^4 \text{ cu. units} \quad \text{1 mark}$	

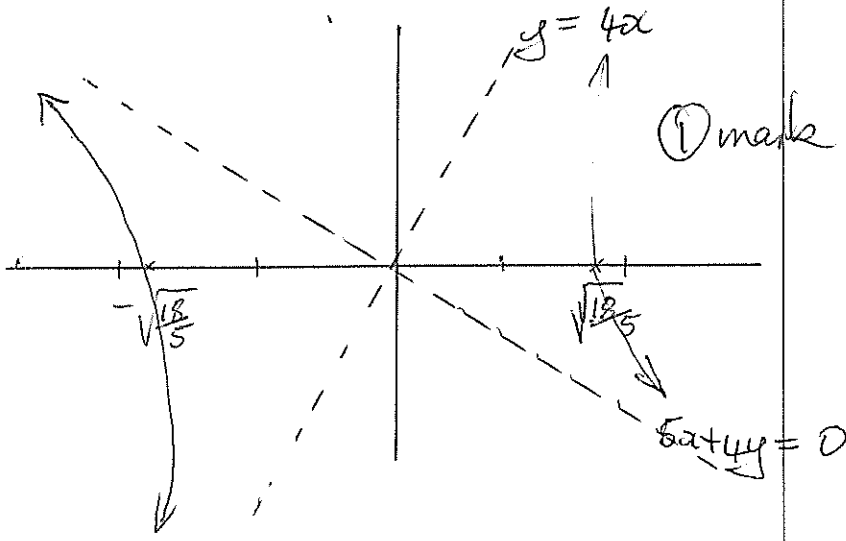
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Suggested solution(s)	comments
Quest 14 (a)(i) $z^5 = -1 = \text{cis}(\pi + 2\pi k)$ $z = \text{cis}\left(\frac{\pi + 2\pi k}{5}\right)$ (1 mark) $k=0 \quad z = \text{cis}\frac{\pi}{5}$ $k=1 \quad z = \text{cis}\frac{3\pi}{5}$ $k=-1 \quad z = \text{cis}-\frac{\pi}{5}$ $k=2 \quad z = \text{cis}\pi$ $k=-2 \quad z = \text{cis}-\frac{3\pi}{5}$ (1 mark)  (1 mark)	
(b) $I_n = \int_0^1 x^{2n+1} e^{x^2} dx$ (1 mark) $u = x^{2n}, \quad v = x e^{x^2}$ $u' = 2n x^{2n-1}, \quad v' = \frac{1}{2} e^{x^2}$ $I_n = \left[\frac{1}{2} x^{2n} \cdot e^{x^2} \right]_0^1 - \int_0^1 n x^{2n-1} e^{x^2} dx$ $= \frac{1}{2} e - n \int_0^1 x^{2n-1} e^{x^2} dx$ (1 mark) $I_n = \frac{e}{2} - n I_{n-1}$	

Suggested Solutions, Marking Scheme and Markers' comments

Suggested solution(s) Quest 14	comments
b (ii) $I_2 = \frac{e}{2} - 2I_1$ $I_1 = \frac{e}{2} - I_0$ ① mark for pattern $I_0 = \int_0^1 x e^{x^2} dx$ $= \left[\frac{1}{2} e^{x^2} \right]_0^1 = \frac{1}{2} e - \frac{1}{2}$ $I_1 = \frac{e}{2} - \frac{e}{2} + \frac{1}{2}$ $I_2 = \frac{e}{2} - 2 \times \frac{1}{2} = \frac{e}{2} - 1$ ① mark	
(c) $5x^2 - y^2 + 4xy = 18$ (i) $10x - 2y \frac{dy}{dx} + 4y + 4x \frac{dy}{dx} = 0$ $\frac{dy}{dx} (4x - 2y) = -10x - 4y$ $\frac{dy}{dx} = \frac{10x + 4y}{2y - 4x} = \frac{5x + 2y}{y - 2x}$ for stat pt $5x + 2y = 0$ ① mark $y = -\frac{5}{2}x$ $5x^2 - \frac{25}{4}x^2 - 4x \times \frac{5}{2}x = 18$ $5x^2 - \frac{25}{4}x^2 - 10x^2 = 18$ ① mark $\frac{real}{no}$ solutions	
(ii) for vert. tangents $y - 2x = 0$ $y = 2x$ ① mark $5x^2 - 4x^2 + 4x \times 2x = 18$ $13x^2 - 4x^2 = 18$ ① mark $x = \pm \sqrt{2}$ $y = \pm 2\sqrt{2}$	

Suggested Solutions, Marking Scheme and Markers' comments

<u>Suggested solution(s)</u> Ques 14	<u>comments</u>
(c)(iii) when $x=0$ $-y^2=18$ $\therefore$ no y intercepts when $y=0$ $5x^2=18$ $x = \pm \sqrt{\frac{18}{5}}$ (iv) for oblique asymptotes $5x - \frac{y^2}{5} + 4y = \frac{18}{y}$ ①mark as $x \rightarrow \infty$ $5x + 4y \Rightarrow 0$ $5x + 4y = 0$ is asymptote $\frac{5x^2}{y} - y + 4x = \frac{18}{y}$ as $y \rightarrow \infty$ $-y + 4x \Rightarrow 0$ $4x - y \Rightarrow 0$ is asymptote ①mark 	

Suggested Solutions, Marking Scheme and Markers' comments

Suggested solution(s) Question 15	comments
(a) $xy = c^2$ $y + x \frac{dy}{dx} = 0$ $\frac{dy}{dx} = -\frac{y}{x}$ (1 mark) at $(cq, \frac{c}{q})$ equation of tang. $\frac{y - \frac{c}{q}}{x - cq} = -\frac{\frac{c}{q}}{cq}$ $\frac{y - \frac{c}{q}}{x - cq} = -\frac{1}{q^2}$ $q^2 y - cq = -x + cq$ $x + q^2 y = 2cq$ (1 mark) (b) $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ $\frac{2x}{a^2} - \frac{2y}{b^2} \frac{dy}{dx} = 0$ $\frac{dy}{dx} = \frac{2x}{a^2} \div \frac{2y}{b^2}$ $= \frac{b^2 x}{a^2 y}$ (1 mark) at $P(x_1, y_1)$ equat of tangent $\frac{y - y_1}{x - x_1} = \frac{b^2 x_1}{a^2 y_1}$ $a^2 y y_1 - a^2 y_1^2 = b^2 x x_1 - b^2 x_1^2$ $b^2 x x_1 - a^2 y y_1 = b^2 x_1^2 - a^2 y_1^2$ $\frac{2x x_1}{a^2} - \frac{y y_1}{b^2} = \frac{x_1^2}{a^2} - \frac{y_1^2}{b^2}$ ie $\frac{2x x_1}{a^2} - \frac{y y_1}{b^2} = 1$ (1 mark)	

Suggested Solutions, Marking Scheme and Markers' comments

Suggested solution(s) 15	comments
(b) similarly tangent thru a $\propto \frac{x_0 x_2}{a^2} - \frac{y_0 y_2}{b^2} = 1$ If $T(x_0, y_0)$ satisfies both equats then $\frac{x_0 x_2}{a^2} - \frac{y_0 y_2}{b^2} = 1$ (1 mark) and $\frac{x_0 x_1}{a^2} - \frac{y_0 y_1}{b^2} = 1$ then $\frac{x x_0}{a^2} - \frac{y y_0}{b^2} = 1$ satisfies P & Q (1 mark) & hence it is chord of contact (iii) If PQ passes thru $S(ae, 0)$ $\frac{ae x_0}{a^2} - 0 = 1$ (1 mark) $x_0 = \frac{a}{e} \text{ which is on directrix}$	
(c) $\ddot{x} = -\frac{k}{x^3}$ (i) $u \uparrow \downarrow g$ when $x = R$ $\ddot{x} = -g$ $-g = -\frac{k}{R^3} \Rightarrow k = gR^3$ (1 mark) (ii) $\ddot{x} = -\frac{gR^3}{x^3}$ $\frac{d(\frac{1}{2}v^2)}{dx} = -gR^3 x^{-3}$ (1 mark) $\frac{1}{2}v^2 = g\frac{R^3}{2}x^{-2} + C$ when $x = R$, $v = u$ $\frac{u^2}{2} = g\frac{R}{2} + C$ $C = \frac{u^2 - gR}{2}$ (1 mark)	

Suggested Solutions, Marking Scheme and Markers' comments

Suggested solution(s) 15	comments
$v^2 = \frac{gR^3}{x^2} + u^2 - gR \quad (1 \text{ mark})$ $v^2 = \frac{gR^3}{x^2} - (gR - u^2)$ (iii) $x = \sqrt{R^2 + 2uRt - (gR - u^2)t^2}$ If $u \geq \sqrt{gR}$ $u^2 \geq gR \leftarrow (1 \text{ mark})$ then $x = \sqrt{R^2 + 2uRt + ct^2}$ where $c \geq 0$ (1 mark) $\therefore x > R$ particle does not return (iv) $v^2 = \frac{gR^3}{x^2} - (gR - u^2)$ when $x = D$ $v = 0$ $0 = \frac{gR^3}{D^2} - (gR - u^2)$ $\frac{gR^3}{D^2} = gR - u^2$ $D^2 = \frac{gR^3}{gR - u^2}$ $D = \sqrt{\frac{gR^3}{gR - u^2}} \quad (1 \text{ mark})$ 2, for time taken $x = R$ $R = \sqrt{R^2 + 2uRt - (gR - u^2)t^2}$ $R^2 = R^2 + 2uRt - (gR - u^2)t^2$ $(gR - u^2)t^2 - 2uRt = 0$ $t[(gR - u^2)t - 2uR] = 0$ $t = 0$ or $t = \frac{2uR}{gR - u^2} \quad (1 \text{ mark})$	

Suggested Solutions, Marking Scheme and Markers' comments

Suggested solution(s) <u>Quest 16</u>	comments
(a) $\omega = 30 \times 2\pi \text{ rad/sec} = 60\pi$ $v = r\omega = 1800\pi \text{ cm/sec.}$ ① mark $a = r\omega^2 = 30 \times (60\pi)^2$ $= 108,000\pi^2 \text{ cm s}^{-2}$ (b) $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ (i) $\frac{2x}{a^2} + \frac{2y}{b^2} \frac{dy}{dx} = 0$ $\frac{dy}{dx} = -\frac{2x}{a^2} \div \frac{2y}{b^2}$ $= -\frac{b^2}{a^2} \frac{x}{y}$ ① mark equat of normal at P $\frac{y - y_0}{x - x_0} = + \frac{y_0}{x_0} \frac{a^2}{b^2}$ ① mark $\therefore y - y_0 = \frac{a^2 y_0}{b^2 x_0} (x - x_0)$ (ii) for $y = 0$ $-y_0 = \frac{a^2}{b^2} \frac{y_0}{x_0} (x - x_0)$ ① mark $-b^2 x_0 = a^2 x - a^2 x_0$ $a^2 x = a^2 x_0 - b^2 x_0$ $x = x_0 \left(\frac{a^2 - b^2}{a^2} \right)$ $x = e^2 x_0$ ① mark	

Suggested Solutions, Marking Scheme and Markers' comments

Suggested solution(s) Quest 16	comments
(ii) $\frac{PS}{PM} = e = \frac{PS'}{PM'}$ ① mark $\frac{PS}{PS'} = \frac{PM}{PM'} = \frac{\frac{a}{e} - x_0}{\frac{a}{e} + x_0}$ $\frac{NS}{NS'} = \frac{ae^2 - e^2 x_0}{ae + e^2 x_0} = \frac{\frac{a}{e} - x_0}{\frac{a}{e} + x_0}$ ① mark $\therefore \frac{PS}{PS'} = \frac{NS}{NS'}$ iv) in $\triangle S'PN$ $\frac{\sin \alpha}{NS'} = \frac{\sin PNs'}{PS'} \quad \text{①}$ in $\triangle SPN$ $\frac{\sin \beta}{NS} = \frac{\sin PNs}{PS} \quad \text{②}$ but $\sin PNs = \sin PNs'$ ① mark $\therefore \sin \alpha \frac{PS'}{NS'} = \sin \beta \frac{PS}{NS}$ $\sin \alpha \frac{PS'}{PS} = \frac{NS'}{NS} \sin \beta \quad \text{① mark}$ i.e. $\sin \alpha = \sin \beta$ from (iii) i.e. $\alpha = \beta$	
(c)(i) $\omega = \frac{2\pi}{24 \times 3600} \doteq 7.3 \times 10^{-5} \text{ rad.s}^{-1}$ ① mark (ii) $F = k \frac{Mm}{x^2}$	

Suggested Solutions, Marking Scheme and Markers' comments

Suggested solution(s)	comments
quest 16 C (ii) (contin) at earth's surface $R = 6400 \text{ km} \quad F = mg$ $mg = \frac{k M m}{R^2} \quad \textcircled{1} \text{ mark}$ $k M = g R^2$ $= 9.8 \times (6400 \times 1000)^2$ $= 9.8 \times (6.4 \times 10^6)^2$ $F = \frac{9.8 \times (6.4 \times 10^6)^2 m}{r^2} \quad \textcircled{1} \text{ mark}$ for satellite in uniform motion $m r \omega^2 = \frac{9.8 \times (6.4 \times 10^6)^2}{r^2}$ (iii) Hence $r^3 = \frac{9.8 \times (6.4 \times 10^6)^2}{(7.3 \times 10^{-5})^2}$ $r \approx 4.22 \times 10^7 \text{ m}$ $= 42232 \text{ km} \quad \textcircled{1} \text{ mark}$ — R. $\approx 36,000 \text{ km}$ (iv) $v = r \omega$ $= 42200 \times 7.3 \times 10^{-5} \text{ km s}^{-1}$ $\approx 3.1 \text{ km s}^{-1}$	